

Vertika Singh

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Education

JSS Academy of Technical Education, Noida, B.Tech in CSE (AI/ML)

Sept 2023 – Oct 2027

- CGPA: 7.5/10.0

Projects

ORBIT- AI Freelance Work Command Center

Live

- Built a production multi-agent system using CrewAI that automates the full freelance business workflow — brief parsing, scope pricing against Indian market rates, proposal drafting, GST-compliant invoice generation, and payment follow-up — reducing admin time from 10+ hours/week to under 30 minutes.
- Designed an event-driven automation layer using Celery + Redis where marking a milestone complete auto-generates an invoice and a daily cron drafts follow-up messages for overdue payments, eliminating all manual triggers.
- Deployed a SaaS architecture with row-level security on PostgreSQL (Supabase), JWT auth, and Next.js — handling complete data isolation across freelancer accounts with zero cross-tenant leakage.
- **Tools Used:** FastAPI, CrewAI, Celery, Redis, PostgreSQL, Supabase, Next.js, TypeScript

Entropy AI- DevOps Log Analyzer

GitHub

- Built a production-style multi-service document/log processing pipeline with Docker Compose (FastAPI API, log-ingestion agent, simulated app, Streamlit dashboard) that detects ERROR/CRITICAL events in real time and triggers automated incident analysis.
- Implemented a RAG-style RCA pipeline that matches errors to DevOps runbooks via keyword retrieval, then uses Groq (Llama 3.1) through LangChain to generate structured, source-grounded root-cause reports from rolling log context.
- Designed an event-driven ingestion flow with a rolling 20-line context window and REST API (POST /api/v1/analyze), plus a live dashboard for log streaming and manual RCA on pasted error snippets.
- **Tools Used:** Python, Groq API, LangChain, Docker, FastAPI

KOKO- Offline AI Agent

GitHub

- Deployed a local AI agent using Ollama (Llama 3B / Moondream 1.8B), ensuring zero data leakage for private system tasks
- Built a custom agentic loop with deterministic routing and regex safety fallbacks to execute Python (psutil) and SQL commands accurately
- Bypassed native JSON formatting limits of small models by engineering an XML-tagging pipeline to structured local SQLite data queries.
- **Tools Used:** Python, Pydantic, Streamlit, Ollama, Llama 3.2, FastAPI, Moondream, SQLite

FemmeCare- Women Health Care Platform

Live

- Engineered a customer-facing MERN-stack telehealth platform connecting patients with gynecologists; implemented JWT-based role-based access control (RBAC) for separate patient and doctor dashboards.
- Integrated Gemini API to power an empathetic AI health assistant and embedded zero-config video consultations via the Jitsi Meet React SDK, eliminating third-party infrastructure costs.
- Designed a node-cron task scheduler for automated email appointment reminders and built real-time slot conflict detection to prevent double-booking across concurrent requests.
- **Tools Used:** Node.js, Express.js, React, Vite, MongoDB, REST APIs, Jitsi Meet

Technical Skills

Languages: Python, JavaScript, Java, C++

CS Fundamentals: Data Structures and Algorithms, OOP, DBMS

ML / AI: TensorFlow, PyTorch, scikit-learn, OpenCV, NLP, RAG, LangChain, LangGraph, Prompt Engineering

GenAI and LLMs: Ollama, LlamaIndex, OpenAI APIs, n8n Automation, CrewAI, Hugging Face

Backend: JavaScript/TypeScript, FastAPI, Node.js, Express.js, Next.js, REST APIs, JWT Authentication, Webhooks

Frontend: HTML, CSS, React

Databases: MongoDB, SQL, PostgreSQL, Supabase

Cloud & DevOps: Docker, Git, GitHub, Google Colab, AWS